Viktoriia Volkova

Designer | Software Engineer | M.Sc. Computational Mathematics @ UJ | ISTCiM Bronze Medalist
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Summary

I am an M.Sc. student in Computational Mathematics at Jagiellonian University with 2.5 years of software-engineering experience, hands-on UX/UI and product design work, and AI model-evaluation experience. I am seeking opportunities in UX/UI Design, Product Design, and AI Design Evaluation.

Experience

Jagiellonian University Computer Science Student Association (KSI UJ) Kraków (hybrid)
Software Engineer Intern (Scalable Backend) *08/2026 (1-month internship)*
- In a four-person team, built from scratch the backend of a scalable online judge: a FastAPI control-plane API with a transactional outbox in PostgreSQL 16, an asynchronous evaluation queue on Redis 7 Streams consumer groups with retries and a dead-letter stream, and a sandboxed execution layer with verdict policy and output checking (Accepted, TLE, Wrong Output).
- Modelled immutable challenge revisions and test cases, idempotent command processing and lifecycle events; defined REST API contracts for the Next.js 15 / React 19 console with a Monaco code editor; ran code reviews and end-to-end module tests.

Promotion Ambassador & Designer *07/2024 – Present*
- Designed 30+ posters and social-media graphics in Figma for 30+ student events, from 40–50-person university meetups to HackYeah (200+ attendees), building a consistent visual identity for KSI UJ; designed the association's T-shirt (20 printed copies).

AI Mathematical Reasoning Evaluator 07/2026 (1-month contract)
Mercor — contract project for xAI *Remote*
- Fixed-term contract: evaluated Grok's solutions to 100+ research-level mathematical problems comparable to Humanity's Last Exam, producing feedback data to improve the model's reasoning; identified reasoning errors, wrote targeted hints and measured how effectively each hint guided the model to a correct solution.

C++ (Qt) Software Engineer 01/2024 – 06/2026
CHM Software *Kharkiv, Ukraine (remote)*
- Refactored POS and self-checkout interfaces from Qt Widgets to QML in C++17 on Linux terminals, cutting UI load time from 2.3 s to 1.1 s (−52%) and improving render smoothness on low-power hardware.
- Implemented asynchronous data updates with Qt Concurrent and QAbstractListModel; modularized legacy CMake builds; improved logging and monitoring during the Qt 6 migration, with Git and CI/CD.

Product & Economy UX/UI Designer 05/2023 – 10/2023
Chit Chat AI *Kraków (remote)*
- Sole web UX/UI designer in an AI startup; designed the web product end-to-end in Figma, from first screens to production-ready designs: landing page, character catalog, responsive chat, sign-up and login flows.
- Designed the in-app economy and gacha UI in Figma: gem shop, limited-time character banners with drop rates, daily streaks and tasks; formulated drop-rate probability distributions, pity/soft-pity logic and virtual-currency sinks and faucets.

Selected Projects

B.Sc. thesis (defended): Visualizing Discrete Morse Theory | *supervisor: Dr. Sławomir Bakalarski* 2026
- Illustrated exposition of discrete Morse theory (Lefschetz complexes, persistence, cancellation).
- Derived and proved a direct $O(n^2k)$ update of a DAG's path-counting matrix after reversing a unique k-edge path, cheaper than full recomputation on dense graphs when $k = o(n)$.

Hybrid CNN–TDA for Prostate Cancer Grading | *Python, PyTorch*
- Six-class ISUP grading pipeline: ResNet50 tile features, attention aggregation and 156 persistent-homology descriptors.

Deffuant Opinion Dynamics | *Python, NumPy, Matplotlib, Jupyter*
- Bounded-confidence model on lattices and vector opinions, with interactive parameter exploration in ipywidgets.

Education

Jagiellonian University — M.Sc. Computational Mathematics 2026 – 2028 (expected)
Relevant coursework: codes and tilings, introduction to game programming, functional analysis *Kraków*

Jagiellonian University — B.Sc. Computational Mathematics (GPA 4.0/5.0) 2023 – 2026
Relevant coursework: computational algebra, math modelling, discrete dynamical systems, logic & set theory *Kraków*

I.O. Dunayevsky School of Arts No. 5 — Certificate with Honors in Fine Arts 2015 – 2021
6-year programme: academic drawing, painting, composition, art history *Kharkiv, Ukraine*

Skills

Design: Figma, UX/UI, product and in-app economy design, data and math visualization, visual and print design
AI & evaluation: LLM output evaluation, reasoning-error analysis, targeted hint writing
Programming: C++17, Qt/QML, QtTest, Python, FastAPI, PostgreSQL, Redis Streams, TypeScript, JavaScript, React, Next.js, Node.js, C, Java, SQL, Bash, CMake, Git/GitLab, Linux, CI/CD
ML & math: PyTorch, torchvision, NumPy, pandas, scikit-learn, OpenCV, Matplotlib, Albumentations, Jupyter; deep learning, computer vision, model training, medical image analysis; persistent homology, discrete Morse theory
Languages: English (C1), Polish (C1, certified), Ukrainian (native), Russian (native)

Achievements & Activities

- **Bronze medal and team captain**, ISTCiM 2025 (International Student Team Competition in Mathematics).
- 2025: XXVI International Workshops for Young Mathematicians: Combinatorics; Polish National Mathematics Student Conference; volunteer at the Student Festival of Computer Science.

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